

NAZIRU HALILU

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[GitHub](#) | [ORCID: 0009-0006-2324-9599](#) | [Zenodo records](#) | [LinkedIn](#)

RESEARCH INTERESTS

Computer vision and geospatial machine learning for digital agriculture; self-supervised and contrastive learning for agronomic imaging; radiometric harmonization and vegetation indexing from UAV and satellite imagery; geostatistics and precision agriculture mapping (regression kriging, singularity indices); hybrid physics-informed deep learning for hydrological and environmental process modelling; closed-loop sensing and control for smart irrigation and greenhouse automation.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD), Cutting-Edge Technologies for Sustainable Agriculture 2025 to expected 2027

Public University of Navarre, Spain · UTAD, Portugal · Agricultural University of Athens, Greece

Bachelor of Engineering (B.Eng.), Agricultural and Bioresources Engineering, First Class Honours 2019 to 2024

Ahmadu Bello University, Zaria, Nigeria · CGPA 4.77/5.00, top 1% of graduating cohort

National Diploma, Agricultural Engineering Technology, Distinction 2016 to 2018

Ahmadu Bello University, Zaria, Nigeria · CGPA 3.78/4.00, top 1% of graduating cohort

PUBLICATIONS

Peer-Reviewed

- **Halilu, N.** (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. *International Journal of Research and Scientific Innovation (IJRSI)*, 13(7), 2920–2950. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Real-Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. *IJRSI*, 13(7), 2951–2989. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. *IJRSI*, 13(7), 2990–3020. [DOI](#)
- Ismail, K., **Halilu, N.** (2024). AI-Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. *Proc. African Conf. on Precision Agriculture (AfCPA)*, 2(1), 100–120.
- Auwal, M., Yusuf, A., **Halilu, N.** (2020). Physio-chemical and bacteriological quality of water for irrigation purpose (A case study of Shika Dam). *Nigerian Research Journal of Engineering and Environmental Sciences*, 5(1), 198–204.

Submitted / Under Review

- **Halilu, N.,** Sulaiman, J. Geometry-Agnostic Contrastive Learning (GACL): A Theoretical Framework and Baseline Evaluation Protocol for Agronomic Imaging. *Biosystems Engineering* (Elsevier). Under review. [DOI](#) | [Code](#)
- **Halilu, N.** (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory (*Ipomoea* spp.). *Environmental Monitoring and Assessment* (Springer Nature). Under review. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. *European Journal of Agronomy* (Elsevier). Under review. [DOI](#) | [Code](#)

Preprints

- **Halilu, N.** (2026). Multitask Singularity Regression Kriging for Spatio-Temporal Soil NPK Crop Stress Yield-Prediction. *Zenodo Preprint* (v1.0.1). [DOI](#) | [Code](#)
- **Halilu, N.** (2026). STADOS: Spectral-Temporal Adaptive Dark Object Subtraction for Radiometrically Harmonized Multi-Temporal Agricultural Vegetation Monitoring. *Zenodo*. [DOI](#) | [Code](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time-Series Modelling and Control Systems in Precision Irrigation. *Zenodo*. [DOI](#)

RESEARCH EXPERIENCE

EMJMD Research Scholar — Public University of Navarre, Spain

Sep 2025 to Jan 2026

Term 1: Technological bases for agricultural sustainability

- Designed STADOS, a five-stage radiometric harmonization framework, cutting temporal NDVI variance by 39% and 31% on real Sentinel-2 imagery over Portuguese corn parcels.
- Built a 14-module open-source RGB/multispectral vegetation indexing and GLCM texture analysis pipeline, reaching 33.8% balanced accuracy against a 4.55% chance baseline.
- Built a Bayesian-grounded digital twin combining hydrological, hydraulic, and hybrid CNN-BiLSTM-attention sediment models for gully erosion biocontrol (hold-out R-squared = 0.90).

EMJMD Research Scholar — UTAD, Portugal

Feb 2026 to Jun 2026

Term 2: Precision agriculture for sustainable crop production

- Developed Multitask Singularity Regression Kriging (MSRK), reducing residual kriging uncertainty by roughly 20% across 17,864 soil samples.
- Integrated GIS, remote sensing, and ML pipelines across three cropping systems, achieving R-squared between 0.978 and 0.995 and a 2.69-tonne reduction in lime demand.

Summer Intern, Computer Vision Group — Ahmadu Bello University

Jun 2026 to Aug 2026

Dept. of Computer Engineering

- Extended the GACL framework with four new geometry-invariant components, reaching 64.4% accuracy (macro AUC 0.882) across 19 disease classes.

Project Assistant (Remote Data Lead), Smart Greenhouse Automation — IAR-ABU, Nigeria

2024 to 2025

- Automated greenhouse environmental control by engineering adaptive irrigation-scheduling algorithms in Python fusing real-time soil-moisture and microclimate telemetry.

Student Research Assistant, Precision Crop Systems — NAERLS, Nigeria

2022 to 2024

- Trained and validated CNN-based crop-stress detection models in TensorFlow, reaching 78% classification accuracy on multispectral UAV imagery.

Project Assistant, Smart Irrigation & Water Systems — BOMO-Basin Project, Nigeria

2021 to 2022

- Automated water delivery across a one-hectare smart drip-irrigation array by implementing closed-loop control logic.

Research Intern — NCAM, Ilorin and IAR, Zaria, Nigeria

2017 to 2020

- Built Python data-logging pipelines benchmarking mechanization efficiency; supported baseline hydrological research.

Founder, Nhalilu's Biogas Plant and Sales Company

2022

Registered with SMEDAN, Nigeria · Bioprocess Engineering

- Founded and registered a biogas production and sales venture applying bioprocess engineering principles to a working commercial operation.

AWARDS AND HONORS

- Erasmus Mundus Joint Master Scholarship, European Union (2025)
- Best Project Award, ABU Senate (2024)
- Best Graduating Student, Agricultural & Bioresources Engineering (2024)
- Nigerian Federal Government Award for Academic Excellence (2024)
- Best Undergraduate Research Paper Award, ABU (2024)
- PTDF Undergraduate Fellow (2023)

TEACHING AND MENTORSHIP

- Undergraduate Course Tutor, ABU Zaria (NAMET), 2020–2024: Differential Equations & Transforms, Irrigation System Design, Computer Applications in Agric. Engineering.
- Deputy Coordinator, KOJYAF, 2020–2025: Mentorship programs preparing secondary students for JAMB entrance examinations.
- Virtual Secondary School Teacher, 2024–present: Remote instruction in Physics and Mathematics.

PROFESSIONAL MEMBERSHIPS

American Society of Agricultural and Biological Engineers (ASABE) · Nigerian Institution of Agricultural Engineers (NIAE) · Nigerian Society of Engineers (NSE) · Pupil Engineer, COREN

REFERENCES

Prof. Javier Casalí Sarasibar — Department of Agricultural Engineering and Bioscience, Public University of Navarre, Spain · jcs@unavarra.es

Prof. Konstantinos Soulis — Agricultural University of Athens, Greece · soco@aua.gr

Dr. Ibrahim Babangida Dalha — Ahmadu Bello University, Nigeria · ibdalha@abu.edu.ng

RESEARCH PROJECTS AND OPEN-SOURCE REPOSITORIES

- Geometry-Agnostic Contrastive Learning (GACL) — <https://github.com/halilunaziru73-creator/Geometry-Agnostic-Contrastive-Learning-GACL>
- STADOS: Spectral-Temporal Adaptive Dark Object Subtraction — <https://github.com/halilunaziru73-creator/STADOS-Spectral-Temporal-Adaptive-Dark-Object-Subtraction>
- Multitask Singularity Regression Kriging (MSRK) — <https://github.com/halilunaziru73-creator/Multitask-Singularity-Regression-Kriging-for-Spatio-Temporal-Soil-NPK-Crop-Stress-Yield-Prediction>
- Digital Twin for Gully Biocontrol Using Morning Glory — <https://github.com/halilunaziru73-creator/Digital-Twin-for-the-Evaluation-of-Experimental-Gully-Biocontrol-Using-Morning-Glory-Ipomoea-spp>
- Hybrid CNN-BiLSTM-Attention Sediment Transport Framework — <https://github.com/halilunaziru73-creator/Hybrid-CNN-BiLSTM-Attention-Sediment-Transport-Agricultural-Gully-System>
- Real-Time RGB Proxy Vegetation Indexing and Texture Analysis — <https://github.com/halilunaziru73-creator/Real-Time-RGB-Proxy-Vegetation-Indexing-and-Texture-Analysis-for-UAV-and-Handheld-Crop-Imagery>
- Operationalizing GIS and ML across Contrasting Cropping Systems — <https://github.com/halilunaziru73-creator/Operationalizing-GIS-and-Machine-Learning-across-Contrasting-Cropping-Systems>
- Geospatial Data Analysis (six-notebook GIS/remote sensing course) — <https://github.com/halilunaziru73-creator/Geospatial-data-analysis>
- GIS-Based Livestock Slurry Suitability Delineation — <https://github.com/halilunaziru73-creator/GIS-based-delineation-of-areas-suitable-for-livestock-slurry-application>
- GIS Monitoring of Grape Moth (*Lobesia botrana*) — <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Grape-Moth-Vineyard-Pest>
- GIS Monitoring of Vineyard Ripening (Brix) — <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Vineyard-Ripening-Brix>
- NaCROP: Crop ET and Irrigation Decision-Support Pipeline — <https://github.com/halilunaziru73-creator/NaCROP>
- Integrated Decision-Support Model for Dairy Processing — <https://github.com/halilunaziru73-creator/Integrated-Decision-Support-Model-for-Dairy-Processing-under-Milk-and-Market-Variability>
- Full repository index — <https://github.com/halilunaziru73-creator>

CONFERENCES, SEMINARS AND WORKSHOPS

- Data Management Workshop: From Phenotyping Experiment to FAIR Data, Norway (2026)
- Digital Agroecology Cluster Conference (EU-funded), Brussels (2025)
- 6th African Regional Conference on Irrigation and Drainage, ARCID (2025)
- Vibrational Spectroscopy & Chemometrics in Agro-Food Systems (EU-funded) (2025)
- Mechanistic Modeling of Rainfall-Driven Soil Solute Transport, University of Florida (2025)
- IEEE Intl. Conf. on Smart Agriculture Technologies (AgriTech), Valencia, Spain (2024)
- NAERLS Technical Seminar Series: AI in the Food Value Chain (2023)

CERTIFICATIONS AND DOCUMENTED RECOGNITIONS

Honorary Awards

- Award of Merit, Senator (SRC/SUG), Samaru College of Agriculture SUG (2017/2018)
- Certificate of Merit, Best 200 Level Student, Dept. of Agric. & Bio-Resources Engineering, ABU (2021)
- Best Overall Graduating Student, National Diploma, Samaru College of Agriculture (2018/2019)

Training

- Certificate of Continuing Professional Development, ICCP, Envirofly Consulting UK (2024)
- Tony Elumelu Foundation Certificate, Business Management Training (2020)
- Genomics and Bioinformatics Bootcamp and short courses (4 certificates), Genomac Institute Inc., USA (2024–2025)
- SIWES Industrial Attachment Attestations, NAERLS and IAR, ABU (2017–2023)
- Mechanistic Modeling of Rainfall-Driven Soil Solute Transport, ISFOOD/UPNA seminar (2025)

Recognition

- ASABE Member Certificate, Member No. 1061249 (2022)
- NIAE Student Member Certificate, Membership No. S2644 (2024)
- Conference Paper Acceptance, 6th ARCID (2025)
- IJRSI Publication Certificates (3), Vol. 13 Issue 7 (2026)
- SMEDAN Certificate of Registration, N.halilus Biogas Plant & Sells (2020)
- Attestation to Proficiency in English Language, ABU (2024)

West African Examinations Council

- West African Senior School Certificate (WAEC), Knowledge Is Power Secondary School, Zaria — 9 subjects including A1 in Mathematics and Geography (Jun 2015)

FULL ACADEMIC TRANSCRIPT RECORD

Complete course-by-course record across National Diploma, B.Eng., and EMJMD programmes, with individually scored marks, is maintained on the candidate's portfolio site under Full Academic Record, and is available on request as an authenticated transcript, legalized by the Federal Ministry of Education, Nigeria and the Consulado General de España en Lagos. [View online](#)

PLACES WORKED AND STUDIED

Nigeria (Zaria) · Spain (Pamplona, Madrid, Barcelona) · Portugal · Greece · Qatar · Morocco · Ethiopia · France · Italy · Germany, across research travel, conferences, and the Erasmus Mundus programme.

RECENT UPDATES

- Sep 2026: Published GIS monitoring datasets for grape moth and vineyard ripening (Brix), archived on Zenodo with DOIs.
- Aug 2026: Two papers accepted at IJRSI, covering sediment transport modelling and real-time vegetation indexing.
- Aug 2026: Archived STADOS on Zenodo, alongside updated releases of the Zaria irrigation DSS and NaCROP.
- Mar 2025: Paper accepted for presentation at the 6th ARCID, Abuja, on sediment transport rate prediction.

GALLERY AND PORTFOLIO LINKS

Field and campus photographs: [View gallery](#) | Full interactive portfolio: halilunaziru73-creator.github.io | GitHub: github.com/halilunaziru73-creator